M.TECH THESIS — HYPERSPECTRAL IMAGE PROCESSING

VMamba: Progressive State-Space Model Architecture for Hyperspectral Image Pansharpening

A Five-Version Evolutionary Study: Spatial Efficiency · Spectral Fidelity · Parallel Scan Algorithms

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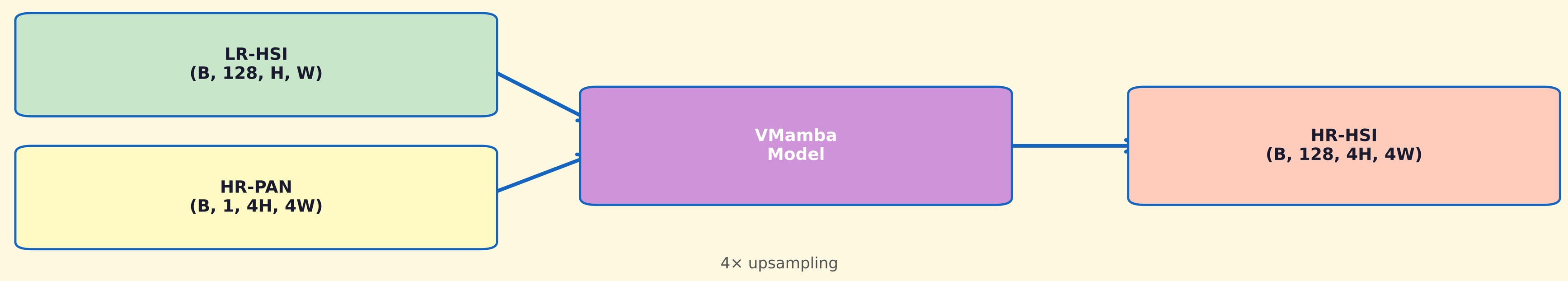
Introduction

- Hyperspectral pansharpening fuses two complementary sensors:
  - LR-HSI : low spatial res, 128 spectral bands (8×8 pixels)
  - HR-PAN : high spatial res, single panchromatic band (32×32)
  - Goal : produce HR-HSI (32×32, 128 bands) at 4× scale
- Challenge: Spatial detail injection must NOT distort spectral signatures — measured by SAM (lower = better).
- Mamba / VMamba State-Space Models offer:
  - O(N) sequence modeling vs O(N²) cross-attention
  - Selective state propagation (data-gated SSM)
  - Efficient 2D spatial scanning (row + column scans)
- This work evolves VMamba through 5 versions, each targeting a specific bottleneck in quality or efficiency.

Problem Formulation & Dataset



- Dataset: Chikusei (Japanese airborne VNIR)
  - 2517 × 2335 pixels · 128 spectral bands
  - Wavelength: ~363–1018 nm
- Patch training: 32×32 HR-HSI patches, downsampled 4× to 8×8 LR-HSI inputs

Architecture Evolution: V1 → V5

V1 Baseline 1.84M	V2 Improved 0.67M	V3 Spectral 1.52M	V4 Best 1.53M	V5 Parallel 1.77M
PSNR 44.65 dB	PSNR 44.99 dB	PSNR 47.35 dB	PSNR 49.37 dB	PSNR 43.34 dB

V1	CrossAttention (O(N²)) + PixelShuffle upsampling	Baseline reference
V2	CrossMamba (O(N)) + Learnable edge injection	−62% params, no checkerboard
V3	SpectralMambaBlock + 4-stage U-Net backbone	+2.36 dB PSNR over V2
V4	Scaled SDIM (α) + StrongReconHead (β) + SpectralGradLoss	Best: PSNR 49.37, ERGAS 4.49
V5	Hillis-Steele O(log L) true parallel scan	6.4× fewer scan iterations

Key Technical Innovations

V4 — Scaled Spatial Detail Injection (SDIM)

```
F_fused = F_hsi + α × (F_pan − F_hsi)
```

α: learnable scalar, initialised at 0.1

- Controls PAN injection strength adaptively
- Prevents spectral distortion from over-injection
- β scalar in Recon Head anchors bicubic baseline

V5 — Hillis-Steele True Parallel Scan

```
Classic SSM scan:  O(L) depth — 32 loops for L=32
Hillis-Steele:      O(log L) depth — 5 loops for L=32

Each pass d with stride = 2^d:
  a_new[t] = a[t] · a[t − stride]
  b_new[t] = a[t] · b[t − stride] + b[t]
```

After log₂(L) passes → all h_t computed in parallel

Result: 6.4× fewer Python iterations vs sequential

Quantitative Results — Chikusei Dataset (Scale 4×)

Version	PSNR (dB) ↑	SAM (°) ↓	ERGAS ↓	SSIM ↑	Params	
V1 (Baseline)	44.65	6.74	71.72	—	1.84M	
V2 (Improved)	44.99	6.77	113.05	—	0.67M	
V3 (Spectral)	47.35	6.61	61.20	—	1.52M	
V4 ★ Best	**49.37**	**6.32**	**4.49**	**0.9997**	**1.53M**	**← BEST**
V5 (Parallel)	43.34	6.74	5.61	0.9622	1.77M	

V4 achieves 13× lower ERGAS than V1 (71.72 → 4.49) with near-perfect SSIM 0.9997
V5 maintains low ERGAS (5.61) while enabling O(log L) parallel training

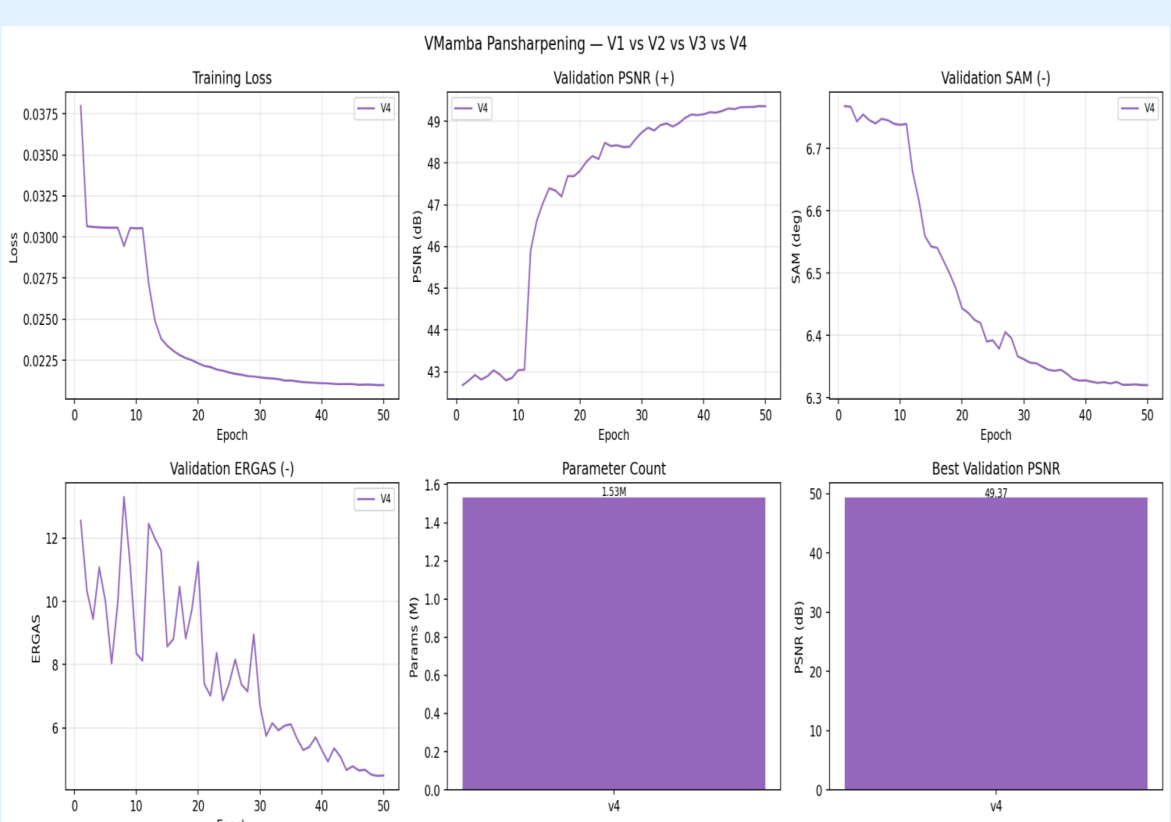


Fig. 1 — V4 Model Comparison (PSNR · SAM · ERGAS · Parameters)

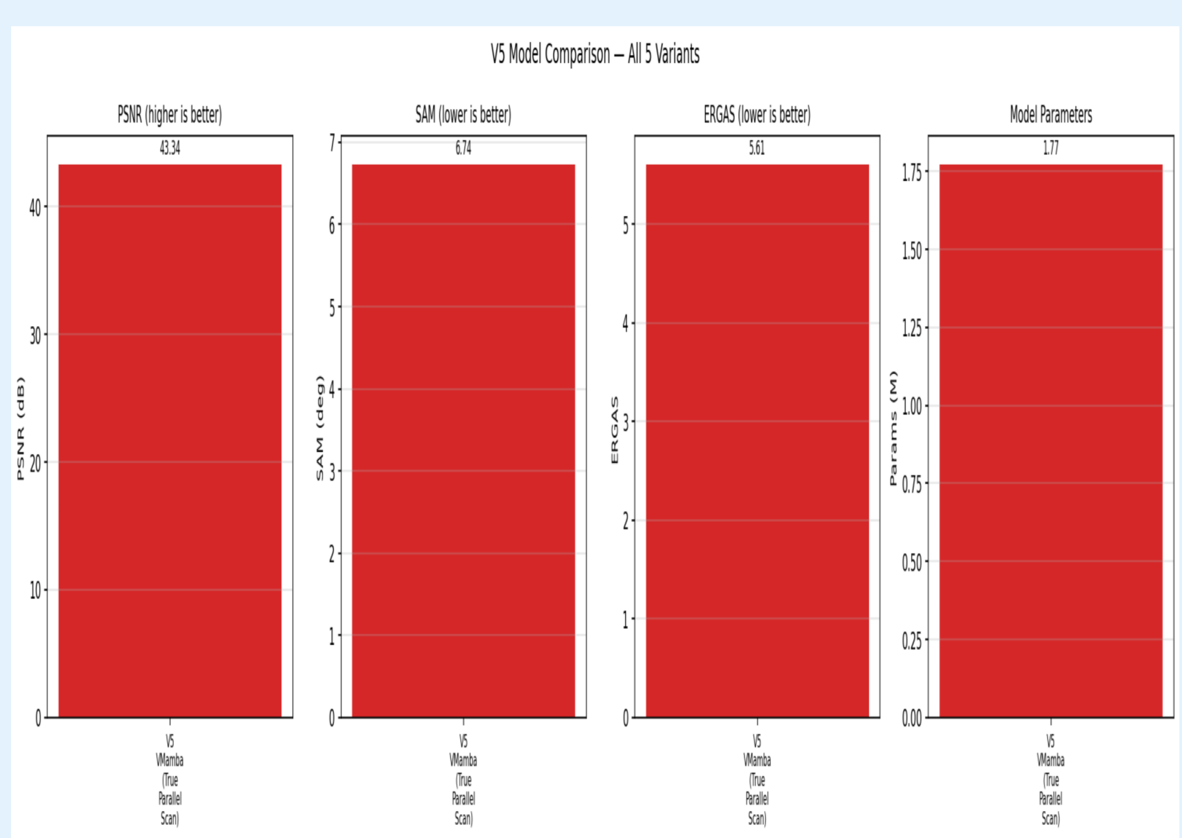


Fig. 2 — V5 Model Comparison (All 5 Variants)

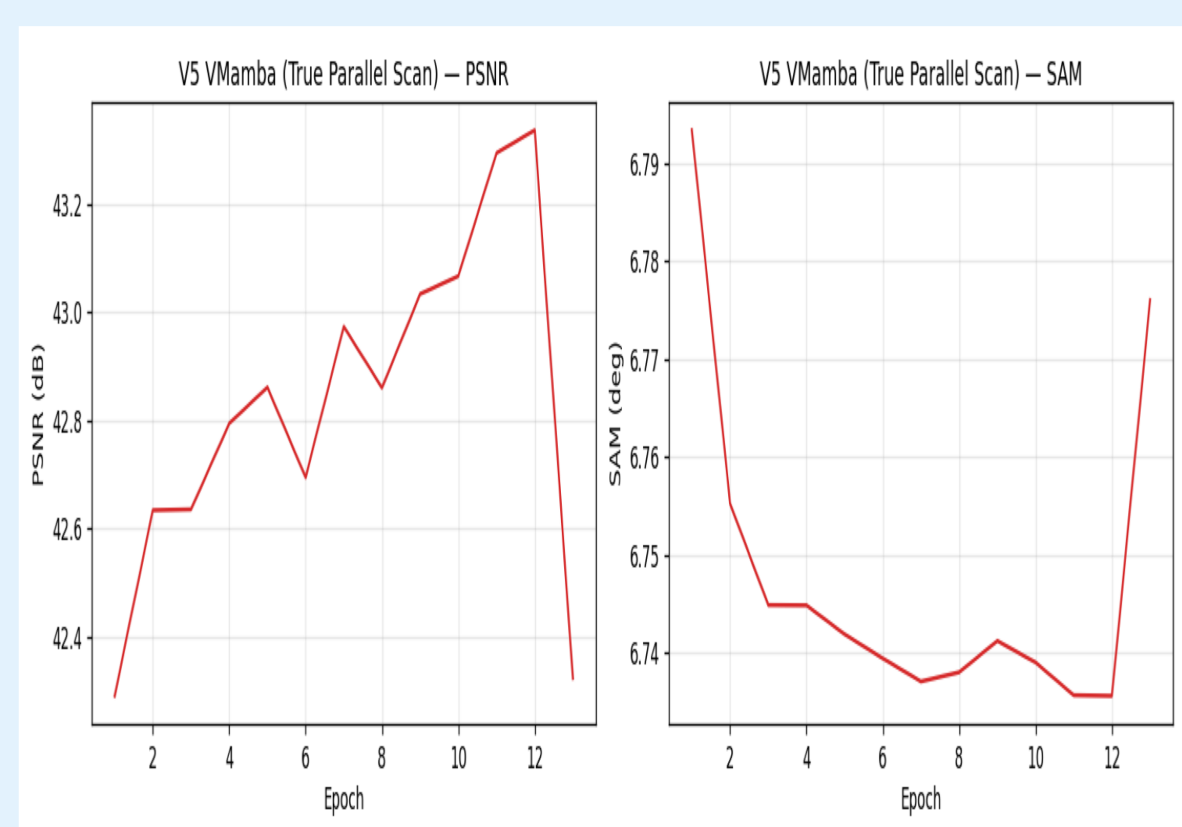


Fig. 3 — V5 Training Curve (PSNR & SAM vs Epoch)

Conclusions

- ✓ Progressive architecture evolution provides a clean ablation study across 5 versions
- ✓ V4 achieves the best spectral fidelity with PSNR 49.37 dB and ERGAS 4.49
  — a 13× improvement in ERGAS over the V1 baseline (71.72 → 4.49)
- ✓ Learnable α (SDIM) and β (Recon Head) scalars are critical for spectral control
- ✓ SpectralGradientLoss penalises band-to-band gradient mismatches, reducing SAM
- ✓ V5 Hillis-Steele scan achieves O(log L) depth: 6.4× fewer loop iterations (L=32)
- ✓ Pre-norm architecture in V5 improves gradient flow and training stability
- ✓ All versions validated on the Chikusei VNIR dataset at 4× upsampling scale

References

[1] Gu et al., "Mamba: Linear-Time Sequence Modeling with Selective State Spaces," arXiv:2312.00752, 2023.

[2] Liu et al., "VMamba: Visual State Space Model," arXiv:2401.13260, 2024.

[3] Yokoya et al., "Airborne Hyperspectral Data over Chikusei," Space Appl. Lab., Univ. Tokyo, Japan, SAL-2016-05-27, 2016.

[4] Vivone et al., "A Critical Comparison Among Pansharpening Algorithms," IEEE TGRS, 2015.

[5] Hillis & Steele, "Data Parallel Algorithms," Comm. ACM, 1986.